



Original research

An isometric neck strengthening program does not improve neck strength in elite women's football-code athletes: A randomised controlled trial

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ABSTRACT

Objectives: The purpose of this study was to investigate the effectiveness of an isometric neck strengthening program to improve isometric neck strength in elite women's football-code athletes.

Design: Randomised controlled trial.

Methods: Elite female soccer ($n = 10$) and Australian football ($n = 30$) players were randomised into either a control ($n = 20$) or experimental ($n = 20$) group for a 12-week intervention study during their respective seasons. While both groups undertook their prescribed strength and conditioning programs, the experimental group also performed isometric neck strengthening exercises three times per week prior to training. Isometric neck strength of the extensors, flexors, lateral flexors, and rotators were assessed pre, mid (Week 7), and post (Week 13) intervention with a hand-held dynamometer during early to mid-competition season. A mixed design analysis of variance was performed for statistical analysis.

Results: No significant group-by-time interactions in isometric neck strength were observed. All strength variables displayed a significant change over time throughout the 12-week period ($p < 0.05$). No significant between group differences in isometric neck strength variables were observed except for lateral left flexion ($F(1, 38) = 5.064$, $p = 0.030$, $\eta_p^2 = 0.117$).

Conclusions: The addition of isometric neck strengthening exercises did not improve neck strength beyond a standard strength and conditioning program for elite women's football-code athletes. While this specific program may not improve neck strength in elite women's football-code athletes, further investigation is needed to determine whether sport-specific neck strength exercises may improve neck strength or if lower-level competition athletes may still benefit from an isometric neck strengthening program.

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Practical implications

- Characterisation of neck strength in elite women's football-code athletes provides comparative data to assist in the identification of individuals who may have low neck strength and are in need of targeted training.
- The addition of specific isometric neck strength training to usual strength and conditioning does not enhance neck strength in elite female football-code athletes
- Isometric neck strength may be improved by undergoing a structured strength and conditioning program.

1. Introduction

Elite athletes that are frequently involved in contact situations such as tackling and/or charging that contributes to a high incidence of

injury.¹ Sport-related concussion (SRC) is a common injury that occurs in collision sports such as football-codes like soccer and Australian football, usually caused by a direct or indirect impact to the head leading to transient functional disturbances to the brain.¹ The head accelerations that occur as a result of impacts are thought to contribute to the attainment of a SRC, with greater accelerations occurring in individuals with weaker neck muscles and smaller neck girths.² As such, individuals with smaller or weaker necks may be at a greater risk of a SRC. While the anatomy of the head and neck structure is not modifiable, neck strength and girth (as a result of increased musculature), as well as muscle activation, can potentially reduce SRC risk.³

Greater neck strength results in greater tensile stiffness and cross-sectional area of the muscles, allowing for more effective absorption of external forces from impacts,⁴ as well as greater contraction speed, allowing for better positioning and bracing of the head-neck segment prior to impact.³ As such, greater neck strength subsequently decreases the magnitude of the head's kinematic response to impact.^{5,6} It has been suggested that isometric strength training of the neck is the most promising training method to activate and strengthen the stabilisation

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muscles of the head-neck segment.⁷ While there is currently insufficient evidence to clearly support the relationship between neck strength and SRC risk,¹ previous research has reported that for every pound (~0.45 kg) increase in neck strength in the cervical muscles, the risk of sustaining a SRC was reduced by 5% in a male and female high school athlete population.⁸ In elite male rugby players, a 5-week isometric neck strengthening program that used minimal equipment was capable of improving neck strength by at least 15% in both sagittal and frontal planes.⁹ But in elite female athletes, no neck strength program has been assessed that may benefit SRC risk reduction.

The latest injury report for the Australian Football League women's (AFLW) competition has shown a higher concussion incidence rate during 2017–2020 (13.85 per 1000 player hours)¹⁰ compared with the concussion incidence rate in the Australian Football League men's competition during 2011–2019 (7.02 per 1000 player hours).¹¹ Given the greater incidence of SRC in women,^{1,10,11} and the growing participation rates of women in contact-based sports such as the various football codes, the quantification of neck strength and the efficacy of an isometric strengthening program in female athletes is warranted. Within the Activate Program (World Rugby Injury Prevention Exercise Program), a self-assisted isometric neck contraction protocol to promote neck strength training has been developed that is targeted for rugby union male and female athletes.¹² While it has been suggested that isometric neck strengthening programs may improve neck strength,^{8,9,13} no research is currently available on the efficacy of neck strengthening programs for elite female athletes. Although males have been shown to have stronger neck muscles, larger neck girths, and greater muscle stiffness,¹⁴ females have been shown to have greater neck strength endurance.^{14,15} Females may therefore have different responses to isometric neck strength programs such as the Activate program than males. As such, the present study aimed to assess the effectiveness of an accessible (i.e., no use of specialised equipment) 12-week neck strengthening program on isometric neck strength. It was hypothesised that those who participate in a 12-week isometric neck strengthening program will have greater strength improvements compared to those who do not.

2. Methods

This study was a randomised control trial (RCT) that aimed to improve neck strength over a 12-week intervention period. A convenience sample of 40 elite female athletes (age 24.4 ± 5.3 years, height 170.0 ± 5.0 cm, mass 67.2 ± 6.3 kg) from one Australian football team ($n = 30$, AFLW) and one soccer team ($n = 10$, W-League) were recruited for this study. Participants were recruited at the team level and were included in the study if they attended the first neck strength testing session. Data collection was undertaken between December 2019 to February 2020, which coincided with pre-season and/or in-season competition phases for both cohorts. To be included in the study, participants must have been 18 years or older and had not have suffered a severe head, neck, or upper body injury in the past six months. All participants provided written informed consent prior to participating in the study, and ethical approval was granted by the La Trobe University Human Ethics Committee (HEC19510). The study adhered to the guidelines described in the National Statement on Ethical Conduct in Human Research and is reported according to the CONSORT guidelines.¹⁶ A flow chart for this study is presented in Fig. 1.

Participants were randomly allocated into either the experimental or control group ($n = 20$ per group: Australian football, $n = 15$; soccer, $n = 5$) by one member of the research team that was not involved in the data collection process (AP). Researchers involved in data collection were blinded to participants' group allocation. Throughout the 12-week period, all participants completed their team-guided strength and conditioning program. Across both sports, these gym programs aimed to improve speed, power, and muscular strength, and included similar key lifts and relative loads. The experimental group completed

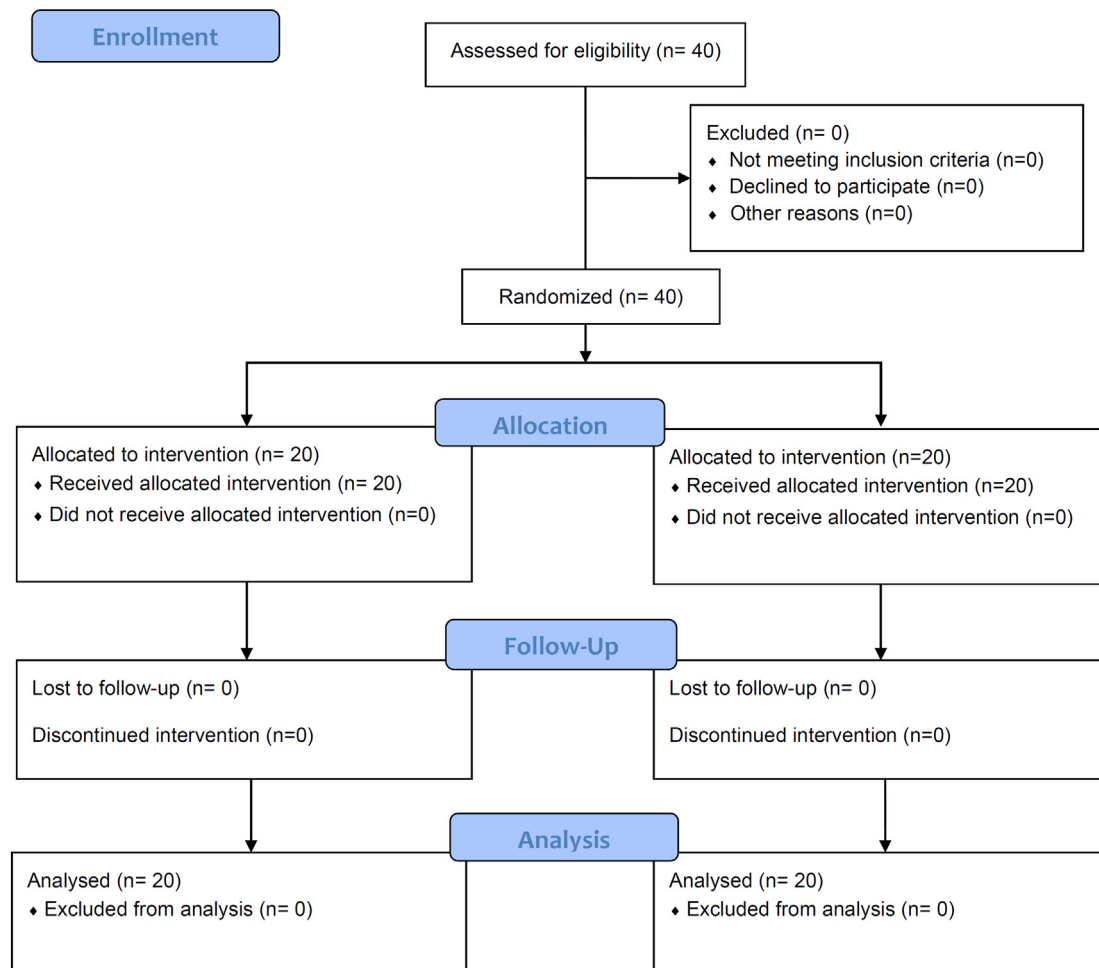
a 12-week pre-activation neck strengthening program in addition to normal field- and gym-based training within the club environment and were monitored using the exercise programming app, *TeamBuildr* and regular weekly check-ins by a member of the research team. The neck exercises were not included in the program for the control group and they continued with the team-prescribed field and gym sessions only. Demographic and player history information was self-reported by participants, including date of birth, years playing in elite competition, years undergoing a structured strength and conditioning (S&C) program, and previous number of concussions. Height and body mass were obtained from previously collected team data.

Assessment of cervical isometric muscle strength was performed in the S&C gymnasium of the players' football club on three separate occasions: pre (week 0), mid (week 7), and post-intervention (week 13). Consideration was taken to ensure testing days followed at least one rest day and 48 h post-match. Isometric neck strength was assessed using a hand-held dynamometer (microFET®2; Hoggan Health, Utah, USA) that measured peak force in Newtons (N). Participants were seated in an upright sitting position in a chair, ensuring that their back was rested on the backrest with both feet firmly on the floor and arms by their side. Participants were instructed to push as hard as they could against the dynamometer for three seconds using only their neck muscles and not their torso. As load was applied onto hand dynamometer, testers encouraged and motivated athletes throughout the assessment of each movement. Two trials for each of the following six movements were taken: flexion, extension, lateral left and right flexion, and 45° oblique left and right rotation. If the recorded peak value for both trials were not within 10% of each other, a third trial was completed for that movement. The average of trials for each movement was used for analysis. The microFET®2 hand-held dynamometer has high intra- and inter-session (ICC: 0.87–0.97)¹⁷ reliability when assessing neck strength and high inter-rater reliability (ICC: 0.91–0.97).¹⁸ Head and neck circumference were also measured at each time point (pre, mid, post) using a measuring tape (Lufkin W606PM 6 mm × 2 m Executive Diameter Pocket Tape).

An accessible and self-directed isometric neck strength program was taken from a previous injury prevention program used in female and male rugby union athletes (World Rugby™ Activate program)¹² that involves isometric strength training in four movements: flexion, extension, and left and right lateral flexion. Participants were instructed to use the palm of their hand to self-generate resistance as hard as they could against their head for ten seconds in each direction and repeat three sets of each movement with ten seconds of rest between sets. Participants did not progress the load throughout the 12 weeks. The experimental group completed the neck strength program three times a week as an activation pre-warmup exercise before on-pitch training and off-pitch S&C programs. These exercises were also included on the club noticeboard or *TeamBuildr* as a reminder for participants to complete. Participants were required to tick off the exercises on app and club noticeboards to confirm exercises were completed, if not a member of the research team would follow up to ensure athlete adheres to the program. As all participants were undergoing in-season competitions, both the experimental and control groups were required to continue their on-and-off pitch S&C programs as normal.

Independent samples *t*-tests were used to identify any differences in baseline demographic information (age, height, body mass, head circumference, years in elite competition, years completing structured S&C program) between the control and intervention groups. Self-reported number of previous concussions were compared between groups using fisher's exact test. Minimal detectable change (MDC) was calculated with standard error of measurement (SEM) using baseline scores to assess minimal amount of within-subject variability. A mixed-design analysis of variance (ANOVA) was used to assess changes in dependant variables (absolute strength measures, N; normalised to body mass strength measures, N/kg; and circumference measures, cm) across time (pre, mid, post) and between group (experimental and

CONSORT 2010 Flow Diagram

Fig. 1. CONSORT Flow Diagram of present study.¹⁶

control). Preliminary analysis found no interaction effect of sport, and so was not included as a factor within the analysis. Significance was set at $p < 0.05$. When significance was met, partial eta-squared effect size (η_p^2) (small = 0.01; medium = 0.06; large = 0.14) were used to measure main effects¹⁹ and Tukey post-hoc test was used. Data are presented descriptively as mean $\pm$ standard deviation. All statistical analyses were performed using Jamovi software (version 1.1.9.0, www.jamovi.org).

3. Results

All participants completed the study protocol without any adverse effects. No significant differences in demographic parameters were present between the experimental group and control group (Table 1). The SEM and MDC values using baseline scores were as follows: flexion (SEM 5.0 N, MDC 13.9 N), extension (SEM 8.0 N, MDC 22.1 N), left flexion (SEM 3.9 N, MDC 10.8 N), right flexion (SEM 4.0 N, MDC 11.0 N), left rotation (SEM 4.6 N, MDC 12.8 N) and right rotation (SEM 4.3 N, MDC 11.9 N).

Absolute strength variables (N) displayed no group-by-time interactions ($p > 0.05$). A main effect for group was observed, with the control group being stronger than the experimental group for extension (experimental $112.03 \text{ N} \pm 25.94 \text{ N}$; control $125.50 \text{ N} \pm 23.38 \text{ N}$, $F(1,38) = 5.447$, $p = 0.025$, $\eta_p^2 = 0.125$), left flexion (experimental $89.58 \text{ N} \pm$

19.73 N ; control $100.13 \text{ N} \pm 13.09 \text{ N}$, $F(1,38) = 8.034$, $p = 0.007$, $\eta_p^2 = 0.175$), right flexion (experimental $90.21 \text{ N} \pm 19.90 \text{ N}$; control $100.23 \text{ N} \pm 16.36 \text{ N}$, $F(1,38) = 5.293$, $p = 0.027$, $\eta_p^2 = 0.122$), and right rotation (experimental $90.68 \pm 18.17 \text{ N}$; control $98.42 \pm 13.25 \text{ N}$, $F(1,38) = 4.817$, $p = 0.034$, $\eta_p^2 = 0.113$). A significant main effect for time was observed for all absolute strength variables ($p < 0.05$).

There were no group-by-time interactions for any normalised strength variables ($p > 0.05$). Only left lateral flexion was statistically different between groups ($F(1, 38) = 5.064$, $p = 0.030$, $\eta_p^2 = 0.117$) while all normalised strength variables displayed significant increases over time ($p < 0.05$) (Fig. 2).

Head-to-neck circumference ratio and neck circumference showed no group by time interaction and no between-group effect ($p > 0.05$). Significant changes over time for both groups were shown for head to neck circumference ratio (pre, 1.64 ± 0.08 ; mid, 1.61 ± 0.07 ; post, 1.60 ± 0.07 ; $F(2,76) = 6.785$, $p = 0.002$, $\eta_p^2 = 0.1515$) and neck circumference (pre, $33.74 \pm 1.76 \text{ cm}$; mid, $34.15 \pm 1.43 \text{ cm}$; post, $34.50 \pm 1.65 \text{ cm}$; $F(2,76) = 5.175$, $p = 0.008$, $\eta_p^2 = 0.1199$).

4. Discussion

This study aimed to assess the effectiveness of an additional 12-week isometric neck strengthening program on isometric neck strength compared with usual strength training in an elite female athlete cohort.

Table 1
Participant demographics.

Demographic parameters	Experimental group (n = 20)	Control group (n = 20)	p-Value	Total (n = 40)
Sport				
AFLW (Australian Football)	15	15	–	30
W-League (soccer)	5	5	–	10
Age (years)	24.6 ± 6.3 (19–40)	24.2 ± 4.1 (18–34)	0.36	24.4 ± 5.2 (18–40)
Height (cm)	171.3 ± 6.1 (160.0–188.0)	169.6 ± 4.9 (162.5–181.0)	0.18	170.0 ± 5.0 (160.0–188.0)
Body mass (kg)	67.0 ± 8.5 (47.0–84.5)	66.8 ± 4.2 (59.2–77.3)	0.46	67.2 ± 6.3 (47–84.5)
Head circumference (cm)	55.0 ± 1.5 (52.0–57.4)	55.3 ± 1.3 (52.5–57.2)	0.26	55.1 ± 1.3 (52.0–57.4)
Years in elite competition	3.90 ± 6.62 (0–26)	4.35 ± 3.87 (0–15)	0.39	4.13 ± 5.36 (0–26)
Completing S&C program (years)	4.40 ± 4.69 (0–20)	5.65 ± 3.20 (2–13)	0.18	5.03 ± 4.02 (0–20)
Previous concussions (self-report)				
Yes (n,%)	10 (50%)	9 (45%)	1.00	19
Number of concussions (n)				
1	7	4	0.70	11
2	2	3		5
3	1	1		2
4	0	1		1

Note: AFLW = Australian Football League Women's; S&C = strength and conditioning. Data presented as mean ± SD (range), or proportion (n). Demographic parameters presented for each group separately (and compared with a t-test). Previous concussions analysed via fisher exact test analysis.

While the additional neck strengthening program did not elicit greater improvements in neck strength compared with the control group, there was improved neck strength, neck circumference, and head-to-neck ratio among both groups across the 12-week period. This highlights that a non-specific S&C program and on-field training contributes to the development of neck strength in elite women's football-code athletes over the course of a season.

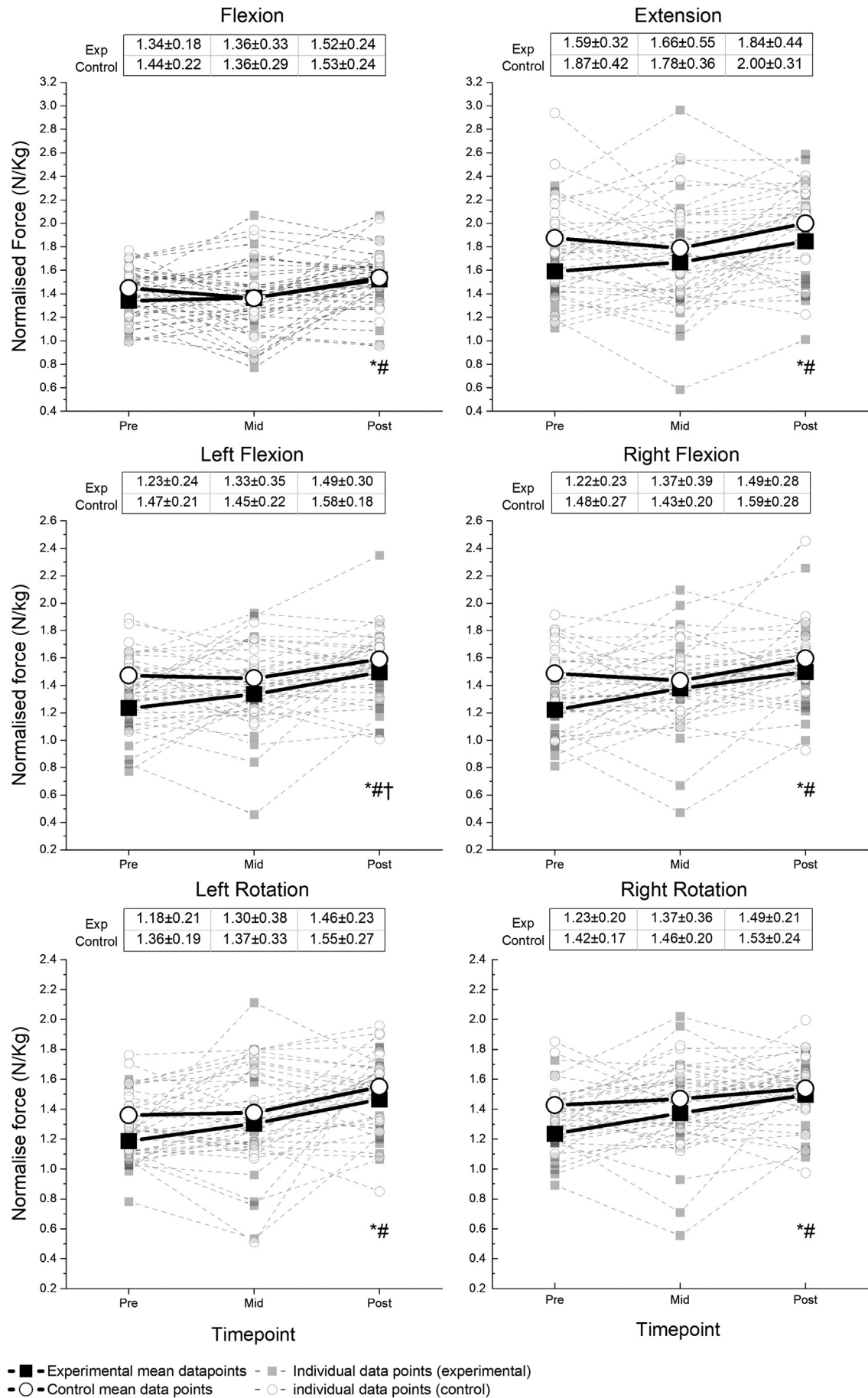
The isometric neck strengthening program used in this study involved the neck strengthening exercises outlined within the recently developed World Rugby Injury Prevention Program 'Activate' (which includes exercises for movement control, pre-activation, and functional conditioning).²⁰ Previous research using this program has reported a 60% reduction in concussion incidence in both schoolboy²¹ and adult men's club²² rugby union players. While the Activate program is targeted for rugby union athletes and the participants in this current study are a mix of female Australian football and soccer players, isolation of just the neck exercises within the program resulted in no significant improvement in neck strength compared to the control group. A positive aspect of these exercises is that they have no specific equipment requirements, allowing it to be accessible to a wide audience and able to be completed in any location. However, given our study showed no effect of these neck strengthening exercises over a 12-week program, it is likely that the reduced concussion incidence reported in these studies^{21,22} may be due to other factors^{23,24} such as improved movement control and technical proficiency within contact situations. More research is therefore required that specifically observes the efficacy of neck strengthening programs and the potential impact these have on reducing the incidence of SRC.

This study showed no improvements in neck strength as a result of this program, which contrasts previous studies assessing the effectiveness of other neck strength programs.^{25–27} Improvements in neck strength from these previous studies have been shown following as little as 5–8 weeks of training.^{9,26,27} Various programs have been used, including the use of isometric exercises,⁹ isoinertial exercises,²⁶ or specialised equipment that self-generates dynamic rotational

resistance.²⁷ However, given that one study found improvements within one week,³ there may be some training effect observed when using unfamiliar equipment (e.g. a neck harness or 4-way neck machine), which may confound the absolute strength improvements that occurred as a result of the program. Indeed, while Geary et al. (2013)⁹ only employed the use of the team S&C coach to create an external force during exercises, other studies have involved the use of specialised equipment.^{25–27} However, while an 8-week program was able to improve isometric neck strength in college American Football male athletes, this did not correspond with a reduction in head accelerations during simulated tackling events.²⁶ In elite male rugby union players, an isometric neck strengthening program elicited non-significant improvements in neck strength after 5-weeks, and over a 26-week season saw no difference in the number of, or time lost due to, cervical spine injuries as a result of the program.²⁵ Research that does show neck strength improvements in elite male athletes appears to require the use of specialist equipment, rather than self-applied resistance.^{25–27} As such, it may be that the elite-level athletes used in this study would also benefit from the use of specialised neck strengthening equipment to elicit further strength gains.

Overall, both groups showed significant improvements in all strength variables from pre- to post-intervention. Likely, this was a result of their team's usual structured S&C programs and/or on-field training requirements to build and maintain general fitness and strength. While not specific to the neck, a number of common gym-based exercises undertaken may elicit secondary strength adaptations in the neck, including the upper trapezius and the sternocleidomastoid muscles.²⁸ As an example, while a deadlift using a trap-bar is typically prescribed to target the lower body, the neck position (and therefore neck muscles) is important to maintain good lifting technique, and the trapezius muscles contribute to the stabilisation of the shoulder throughout the movement.²⁸ Thus, having a structured S&C program for elite women's football-code athletes may be sufficient to strengthen the neck musculature, however, the optimal neck strength of female athletes to reduce the risk of injury is still unknown. Certainly, the

Fig. 2. Normalised data of isometric neck strength (N/kg) throughout the 12-week program (* denotes significant difference pre-post ($p < 0.05$), # denotes significant differences mid-post ($p < 0.05$), †denotes significant group difference).



accessibility of S&C programs are common within the elite environment, but for female athletes involved in lower competition levels such as amateur or community sport clubs, they may not have access to such gym programs or be able to afford S&C coaches. As such, the efficacy of a simple neck strengthening program such as this one may be worth investigating in non-elite athletes. While both groups were able to improve their neck strength throughout the course of this study, whether this was sufficient to meaningfully reduce their SRC risk is unknown. Therefore, the direct link between improved neck strength and reduced SRC risk still needs to be elucidated, and other aspects to reduce the concussion risk in female athletes are still necessary.

Several studies have reported the neck strength of elite male athletes,^{9,25,29} however, no study other than this one has reported on the typical neck strength of female athletes. The only research that quantifies neck strength in women are in individuals with chronic neck pain.^{30,31} Understanding the typical neck strength of elite female athletes may assist in the comparison and identification of women who have weaker necks and may require more specific training to address this weakness. Alternatively, this information may be useful for sub-elite or junior women athletes who want to progress to higher levels of competition to ensure their body is prepared for the demands of the sport. If neck strength is shown to be linked to SRC risk, ensuring that junior athletes have sufficient neck strength will be of particular importance given the potential long term effects on brain development and function following a concussion.³²

There are a number of limitations and considerations that should be taken into account when assessing the generalisability of the present data. Firstly, this study specifically observed the efficacy of a neck strengthening program that was within a larger injury prevention program, originally in rugby union.¹² Whilst our study did not show an improvement in isometric neck strength, potentially, other neck strengthening programs that focus on more dynamic or sport specific movements may be better placed to observe strength improvements and warrant further research to assess suitability across a range of sports and competition levels. Within this study, it was not possible to assess the intent at which participants completed the program. Motivation may influence the load in which they completed these exercises, which would impact the overall efficacy of the program. Similarly, while we did not measure the contribution of soccer heading to neck load in this study, there is a potential that this sport-specific action may also contribute to the strength improvements in soccer players throughout a season. Load from the force applied within these neck exercises, as well as additional load that may contribute from sport-specific movements, should be taken into consideration for future research in this area. Additionally, this study only measured the absolute strength changes following this program. Isometric resistance training has shown to improve rate of force development,³³ and as such, would be a worthwhile metric to observe in future neck strengthening programs.

5. Conclusion

The isometric neck strengthening program utilised in this study did not significantly improve neck strength compared to the control group. However, neck strength over the season was improved in both groups while continuing to undertake their typical sport-specific field and gym training. While the results of this study suggest that this neck strengthening program is not effective for elite women's football-code athletes, further investigation is needed to determine if utilising sport-specific neck strength exercises may improve neck strength. Further investigation is also needed to determine suitability in lower-level competition athletes who may still benefit from an isometric neck strengthening program.

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Declaration of interest statement

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: AJP currently receives partial research salary funding from Sports Health Check charity (Australia) and Erasmus+ strategic partnerships program (2019-1-IE01-KA202-051555).

AJP has previously received partial research funding from the Australian Football League, Impact Technologies Inc., and Samsung Corporation, and has provided expert reports in concussion legal proceedings.

All other authors have no other computing interests to this study.

Confirmation of ethical compliance

All players provided written informed consent prior to participating in the study, and ethical approval was granted by the La Trobe University Human Ethics Committee (HEC19510).

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